

Problem 2.7

Suppose that

$$a(t) = 0.10t^2 + 1.$$

The only investment made is \$300 at time 1. Find the accumulated value of the investment at time 10.

Solution

If an amount is invested at time 1, then its accumulated value at time 10 is obtained by multiplying by the factor

$$\frac{a(10)}{a(1)}.$$

Therefore,

$$\text{Accumulated value at time 10} = 300 \cdot \frac{a(10)}{a(1)}.$$

Now compute:

$$a(10) = 0.10(10)^2 + 1 = 0.10(100) + 1 = 10 + 1 = 11,$$

and

$$a(1) = 0.10(1)^2 + 1 = 0.10 + 1 = 1.1.$$

Thus,

$$300 \cdot \frac{a(10)}{a(1)} = 300 \cdot \frac{11}{1.1} = 300 \cdot 10 = 3000.$$

Hence, the accumulated value of the investment at time 10 is

$$\boxed{\$3000}.$$